

5 hours on examples

Problem 1 (03JPN1)

P is on a circle centered at F with diameter AB so $\angle APB = 90^\circ$. Now, we see that,

$$\angle A = \angle BPA + \angle PAD = 90^\circ - \angle ADB + \angle BAP = 3x - 90^\circ$$

This means,

$$BP = AB \cdot \sin(x) = b$$

We scale things such that the diameter of (BFP) is one unit.

We let $x = \angle FPB = \angle ACF$ and $y = \angle BFP$. Clearly, by law of sines,

$$CP = AF = BF = \sin(x);$$

$$BP = \sin(y);$$

$$FP = \sin(x + y)$$

Now doing law of sines on AFC ,

$$\begin{aligned} 1 + \frac{\sin(x + y)}{\sin(x)} &= \frac{CF}{AF} = \frac{\sin(\angle FAC)}{\sin(\angle ACF)} = \frac{\sin(\angle BEC - \angle ABE)}{\sin x} = \frac{\sin((180 - 2x) - (180 - x - y))}{\sin x} \\ &\Leftrightarrow 1 + \frac{\sin(x + y)}{\sin(x)} = \frac{\sin(y - x)}{\sin(x)} \\ &\Leftrightarrow \sin(x) = \sin(y - x) - \sin(x + y) \\ &\Leftrightarrow 1 = -2 \cos(y) \end{aligned}$$

Clearly, $y = 120^\circ$ is the only possible solution as desired.

Problem 3 (19EGMO3)

We let the intersection of the $\angle BAD$ bisector with BC be P_1 and that of $\angle BXC$ be P_2 .

$$\begin{aligned} \frac{BP_1}{BC} &= \frac{BP_1}{BD} \cdot \frac{BD}{BC} = \frac{AB}{AB + AD} \cdot \frac{a - CD}{a} \\ &= \frac{c}{c + \frac{b}{a} \cdot c} \cdot \frac{a - \frac{b^2}{a}}{a} = \frac{1}{1 + \frac{b}{a}} \cdot \frac{a^2 - b^2}{a^2} = \frac{a}{a + b} \cdot \frac{a^2 - b^2}{a^2} = \frac{a - b}{a} \end{aligned}$$

Equivalently, $\frac{BP_1}{P_1C} = \frac{a-b}{b}$. We now work on $\frac{BP_2}{P_2C} = \frac{BX}{XC}$. Let BA intersect (AIX) again at Y . Note that, looking at $\overline{CAA}$ and $\overline{BYA}$, there is a spiral similarity at X sending BY to CA . Therefore,

$$\frac{BX}{XC} = \frac{BY}{CA} = \frac{BY}{b}$$

We show that $BY = a - b$ to finish. Indeed, since,

$$\angle IYA = \angle IXA = \angle IAC$$

We know that $AI = IY$. Therefore, if we let F be the base of the altitude from I to AB , $AY = 2AE = 2(s - a) = b + c - a$. Thus, $BY = AB - AY = a - b$.

5 clubs; 7 hours

Problem 4 (09IMO4)

We know $\angle B = \angle C = 90 - \frac{\angle A}{2}$. Therefore, $\angle BEC = 45 - \frac{\angle A}{4} + \angle A = 45 + \frac{3\angle A}{4}$. If $\angle IEK = 45$, then we know $\angle KEC = \frac{3\angle A}{4}$. Therefore,

$$\begin{aligned} \frac{IK}{KC} &= \frac{IE}{EC} \cdot \frac{\sin 45}{\sin \frac{3\angle A}{4}} = \frac{\sin \angle ECI}{\sin \angle CIE} \cdot \frac{\sin 45}{\sin \frac{3\angle A}{4}} = \frac{\sin \frac{\angle C}{2}}{\sin \angle BIC} \cdot \frac{\sin 45}{\sin \frac{3\angle A}{4}} = \frac{\sin \frac{\angle C}{2}}{\cos \frac{\angle A}{2}} \cdot \frac{\sin 45}{\sin \frac{3\angle A}{4}} \\ &= \frac{\sin \frac{\angle C}{2}}{\cos \frac{\angle A}{2}} \cdot \frac{\sin 45}{\sin \frac{3\angle A}{4}} = \frac{\sin \left(45 - \frac{\angle A}{4}\right)}{\cos \frac{\angle A}{2}} \cdot \frac{\sin 45}{\sin \frac{3\angle A}{4}} \end{aligned}$$

Also, since $DI \perp DC$, $\frac{IK}{KC} = \frac{DI}{DC} = \tan \frac{C}{2} = \tan \left(45 - \frac{\angle A}{4}\right)$. Therefore,

$$\begin{aligned} \frac{\sin \left(45 - \frac{\angle A}{4}\right)}{\cos \frac{\angle A}{2}} \cdot \frac{\sin 45}{\sin \frac{3\angle A}{4}} &= \tan \left(45 - \frac{\angle A}{4}\right) \\ \Leftrightarrow \cos \left(45 - \frac{\angle A}{4}\right) \cdot \sin(45) &= \cos \left(\frac{\angle A}{2}\right) \cdot \sin \left(\frac{3\angle A}{4}\right) \\ \Leftrightarrow \cos \left(45 - \frac{\angle A}{4}\right) \cdot \sin(45) &= \cos \left(\frac{\angle A}{2}\right) \cdot \sin \left(\frac{3\angle A}{4}\right) \\ \Leftrightarrow \frac{\sin \left(90 - \frac{\angle A}{4}\right) + \sin \left(\frac{\angle A}{4}\right)}{2} &= \frac{\sin \left(\frac{5\angle A}{4}\right) + \sin \left(\frac{\angle A}{4}\right)}{2} \\ \Leftrightarrow \sin \left(90 - \frac{\angle A}{4}\right) &= \sin \left(\frac{5\angle A}{4}\right) \end{aligned}$$

On $0 < \angle A < 180$, the solutions are $\angle A = 60, 90$.

Problem 5 (19SLG2)

We find the inradii of ω_B, ω_C . Note that triangles BFD, BCA are similar. Therefore,

$$r_{BFD} = r_{BCA} \cdot \frac{BF}{BC} = r_{BCA} \cdot \cos B$$

Where R is the circumradius of (ABC) .

$$= 4R \sin\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right) \sin\left(\frac{C}{2}\right) \cdot \cos B$$

Analogously, $r_{CDE} = 4R \sin\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right) \sin\left(\frac{C}{2}\right) \cdot \cos C$. Note that, by law of sines,

$$PM = QN$$

$$\Leftrightarrow 2r_{BFD} \sin FMP = 2r_{CDE} \sin ENQ$$

$$\Leftrightarrow \cos B \sin FMP = \cos C \sin ENQ$$

$$\Leftrightarrow \frac{\sin FMP}{\sin ENQ} = \frac{\cos C}{\cos B}$$

Let N' be where the incircle of ω_C touches BC . Note that,

$$\frac{\sin FMP}{\sin ENQ} = \frac{\sin DMN}{\sin DNM} = \frac{DN}{DM} = \frac{DN'}{DM}$$

Since triangles DBF and DEC are similar, quadrilaterals $DBFM$ and $DECN'$ are similar. Therefore,

$$\frac{DN'}{DM} = \frac{CE}{BF} = \frac{\cos C}{\cos B}$$

As desired.

11 clubs; 8 hours

Problem 7 (13JMO5)

We let $\angle YXB = x$; $\angle BXA = y$. Therefore,

$$\frac{BY}{XP} = \frac{2R \sin x}{\frac{2R \sin(90 - x - y)}{\cos y}} = \frac{\sin x \cos y}{\cos(x + y)}$$

Since $\angle YXC = 90 - \angle AZX = 90 - \angle APX = y$,

$$\frac{CY}{XQ} = \frac{2R \cos y}{\frac{2R \sin(90 - x - y)}{\sin y}} = \frac{\sin y \cos x}{\cos(x + y)}$$

As a result,

$$\frac{BY}{XP} + \frac{CY}{XQ} = \frac{\sin(x + y)}{\cos(x + y)}$$

Indeed, this is $\tan(x + y) = \frac{AY}{AX}$.

Problem 8 (01IMO5)

Simply,

$$AB + BP = AQ + QB$$

$$\Leftrightarrow 2R \sin C + \frac{2R \sin C \sin 30}{\sin(B + 30)} = \frac{2R \sin C \sin\left(\frac{B}{2}\right)}{\sin\left(C + \frac{B}{2}\right)} + \frac{2R \sin C \sin(60)}{\sin\left(C + \frac{B}{2}\right)}$$

$$\Leftrightarrow 1 + \frac{\sin 30}{\sin(B + 30)} = \frac{\sin\left(\frac{B}{2}\right) + \sin(60)}{\sin\left(C + \frac{B}{2}\right)}$$

Note that $B + C = 180 - A = 120$.

$$\begin{aligned} \Leftrightarrow 1 + \frac{\sin 30}{\sin(B + 30)} &= \frac{\sin\left(\frac{B}{2}\right) + \sin(60)}{\sin\left(120 - \frac{B}{2}\right)} \Leftrightarrow 1 + \frac{\sin 30}{\sin(B + 30)} = \frac{\sin\left(\frac{B}{2}\right) + \sin(60)}{\sin\left(60 + \frac{B}{2}\right)} \\ \Leftrightarrow 1 + \frac{\sin 30}{\sin(B + 30)} &= \frac{2 \sin\left(\frac{B}{4} + 30\right) \cos\left(\frac{B}{4} - 30\right)}{2 \sin\left(\frac{B}{4} + 30\right) \cos\left(\frac{B}{4} + 30\right)} \Leftrightarrow 1 + \frac{\sin 30}{\sin(B + 30)} = \frac{\cos\left(\frac{B}{4} - 30\right)}{\cos\left(\frac{B}{4} + 30\right)} \\ \Leftrightarrow \frac{\sin 30}{\sin(B + 30)} &= \frac{\cos\left(\frac{B}{4} - 30\right) - \cos\left(\frac{B}{4} + 30\right)}{\cos\left(\frac{B}{4} + 30\right)} \Leftrightarrow \frac{\sin 30}{\sin(B + 30)} = \frac{2 \sin\left(\frac{B}{4}\right) \sin(30)}{\cos\left(\frac{B}{4} + 30\right)} \\ \Leftrightarrow \cos\left(\frac{B}{4} + 30\right) &= 2 \sin(B + 30) \sin\left(\frac{B}{4}\right) \\ \Leftrightarrow \cos\left(\frac{B}{4} + 30\right) &= \cos\left(-\frac{3B}{4} - 30\right) - \cos\left(\frac{5B}{4} + 30\right) \\ \Leftrightarrow \cos\left(\frac{B}{4} + 30\right) + \cos\left(\frac{5B}{4} + 30\right) &= \cos\left(\frac{3B}{4} + 30\right) \\ \Leftrightarrow 2 \cos\left(\frac{3B}{4} + 30\right) \cos\left(\frac{B}{2}\right) &= \cos\left(\frac{3B}{4} + 30\right) \Leftrightarrow \cos\left(\frac{3B}{4} + 30\right) \left(2 \cos\left(\frac{B}{2}\right) - 1\right) = 0 \end{aligned}$$

We find that the only solution on $B < 120$ is $B = 80$. The angles are $A = 60$; $B = 80$; $C = 40$.

$$\begin{aligned} \Leftrightarrow \cos(50) &= \cos(-90) - \cos(130) \\ \Leftrightarrow \cos(50) &= 2 \sin(110) \sin(20) \\ &= \sin\left(\frac{\sin(-10) - \cos(50)}{\sin(30)}\right) \end{aligned}$$

$$1 + \frac{\sin(30)}{\sin(110)} = \frac{\cos(-10)}{\cos(50)}$$

16 clubs; 10 hours

Problem 10 (19TSTST5)

(Is this the triglen solution? I read each solution in the overleaf and didn't see prominently triglen)

Let P be where ray HK intersects (ABC) again and Q be where ray KH intersects (ABC) . We show that,

$$\begin{aligned} DP \perp BC &\Leftrightarrow DP \parallel AH \Leftrightarrow \frac{PK}{KH} = \frac{DK}{KA} \\ &\Leftrightarrow \frac{PK}{DK} = \frac{KH}{KA} \Leftrightarrow \frac{AK}{KQ} = \frac{KH}{KA} \Leftrightarrow AK^2 = KH \cdot KQ \Leftrightarrow KE^2 = KH \cdot KQ \end{aligned}$$

Claim 1: KE is tangent to (BEH)

Proof: $\angle FEK = 90 - A = \angle HBE$. $\square$

Similarly, KF is tangent to (HFC) . Note that K, H hence both lie on the radical axis of $(BEH), (HFC)$. Therefore, the second intersection of these two circles lies on KH . However, by Miquel's theorem on AEF , their intersection lies on (ABC) meaning it is Q .

$(BEHQ)$ is cyclic. As a result, $KE^2 = KH \cdot KQ$ as desired.

Problem 11 (17SLG1)

We equivalently show that the line through P perpendicular to BC passes through E . Let G be the base of the altitude from P to BC . Let $E_1 = DE \cap GP$; $E_2 = AE \cap GP$. We show that $PE_1 = PE_2$ to finish.

$$AB = BC = CD = 1; \angle B = \angle D = 2y; \angle C = \angle A = 2x$$

By isosceles triangles, $\angle CAB = \angle BCA = 90 - y$; $\angle DBC = \angle CDB = 90 - x$. Angle chasing, we'd get $\angle BPA = 180 - x - y$; $\angle ABP = 2y + x - 90$. Therefore,

$$AP = \frac{\sin(2y + x - 90)}{\sin(180 - x - y)}$$

Looking at APE_2 ,

$$PE_2 = \frac{AP \cdot \sin(\angle PAE_2)}{\sin(\angle AE_2P)} = \frac{AP \cdot \sin(2x + y - 90)}{\sin(270 - 2x - 2y)} = \frac{\sin(2y + x - 90) \cdot \sin(2x + y - 90)}{\sin(180 - x - y) \cdot \sin(270 - 2x - 2y)}$$

This is symmetric in x, y , so when we calculate PE_1 , we will get the same result.

22 clubs; 12 hours

Problem 13 (12SLG3)

Claim: I the incenter of ABC lies on the radical axis of (AI_1C) , (BI_2C) .

Proof: Note that $AI_1 \cap BI_2 = I$. Therefore, it suffices to prove,

$$II_1 \cdot IA = II_2 \cdot IB$$

WLOG, let the inradius of ABC be $r = 1$. Note that, since AEF and ABC are similar triangles, the inradius r_1 of AEF would be $r_1 = r \cdot \frac{AF}{AC} = \cos(A)$. Therefore,

$$AI_1 = \frac{r_1}{\sin\left(\frac{A}{2}\right)} = \frac{\cos(A)}{\sin\left(\frac{A}{2}\right)}$$

Given that $AI = \frac{r}{\sin\left(\frac{A}{2}\right)} = \frac{1}{\sin\left(\frac{A}{2}\right)}$, we have $II_1 = \frac{1 - \cos(A)}{\sin\left(\frac{A}{2}\right)}$. Therefore,

$$\begin{aligned} II_1 \cdot IA &= \frac{1 - \cos(A)}{\sin\left(\frac{A}{2}\right)} \cdot \frac{1}{\sin\left(\frac{A}{2}\right)} = \frac{1 - \cos(A)}{\sin^2\left(\frac{A}{2}\right)} = \frac{1 - \cos(A)}{\sin^2\left(\frac{A}{2}\right)} \\ &= \frac{1 - \cos^2\left(\frac{A}{2}\right) + \sin^2\left(\frac{A}{2}\right)}{\sin^2\left(\frac{A}{2}\right)} = \frac{\cos^2\left(\frac{A}{2}\right) + \sin^2\left(\frac{A}{2}\right) - \cos^2\left(\frac{A}{2}\right) + \sin^2\left(\frac{A}{2}\right)}{\sin^2\left(\frac{A}{2}\right)} = 2 \end{aligned}$$

Analogously,

$$II_2 \cdot IB = \frac{1 - \cos(B)}{\sin^2\left(\frac{B}{2}\right)} = 2 \quad \square$$

Therefore,

$$O_1O_2 \parallel I_1I_2 \Leftrightarrow CI \perp I_1I_2$$

We may angle chase to finish noting that (AI_1I_2B) is cyclic using Theorem 2.9.

Problem 14 (23MEX3)

$$AB \cdot CD = 4PM \cdot PK \Leftrightarrow MB \cdot KC = PM \cdot PK$$

$$\Leftrightarrow \frac{MB}{PK} = \frac{MP}{CK} \Leftrightarrow \Delta BMP \sim \Delta PKC$$

We show this similarity to finish. Note that,

$$\begin{aligned} \frac{BP}{PC} \cdot \sin(\angle BPN) \cdot \sin(\angle NPC) &= \frac{BN}{NC} \Leftrightarrow \frac{BP}{PC} \cdot \sin(\angle BPN) \cdot \sin(\angle NPC) = 1 \\ \Leftrightarrow \frac{BP}{PC} &= \frac{\sin(\angle NPC)}{\sin(\angle BPN)} \Leftrightarrow \frac{BP}{PC} = \frac{\sin(\angle PDA)}{\sin(\angle PAD)} \Leftrightarrow \frac{BP}{PC} = \frac{PA}{PD} \end{aligned}$$

Now note that,

$$\begin{aligned}\angle BPC &= \angle PAD + \angle PDA \Leftrightarrow \angle BPC + \angle APD = 180 \\ \Leftrightarrow \angle BPA + \angle DPC &= 180 \Leftrightarrow \angle BPA = \angle CDP + \angle DCP\end{aligned}$$

Claim: $\angle BPM = \angle KCP$.

Proof: Note that,

$$\frac{BP}{PC} = \frac{PA}{PD} \Leftrightarrow \frac{BP}{PA} = \frac{CP}{PD} \Leftrightarrow \frac{\sin(\angle MPB)}{\sin(\angle APM)} = \frac{\sin(\angle CDP)}{\sin(\angle PCD)}$$

Note that the value $\frac{\sin(\angle CDP)}{\sin(\angle PCD)}$ is fixed whereas $\frac{\sin(\angle M'PA)}{\sin(\angle BPM')}$ is strictly increasing as a point M' "slides" from A to B since the sum $\angle M'PA + \angle M'PB = \angle BPA = \angle CDP + \angle DCP$ is fixed. Therefore,

$$\Leftrightarrow \angle MPB = \angle DCP; \angle APM = \angle PCD \Rightarrow \angle BPM = \angle KCP$$

Symmetrically, we have $\angle CPK = \angle ABP$; $\angle DPK = \angle PAB \Rightarrow \angle CPK = \angle MBP$. Using these two, we have triangles MBP and KPC are similar as desired.

28 clubs; 15 hours

Problem 16 (24USATST2)

Claim: $BP = BQ$.

Note that since $\angle PAB = 90 - \angle BAI = 90 - \frac{A}{2}$, we have,

$$BP = AB \cdot \sin(\angle PAB) = c \cdot \cos\left(\frac{A}{2}\right)$$

On the other hand, we may calculate using $BQ \perp AC$,

$$\angle QBA = 90 - A$$

Using tangency,

$$\angle BAQ = \angle IBQ = \frac{B}{2} - (90 - A) = A + \frac{B}{2} - 90$$

Therefore, $\angle AQB = 180 - \frac{B}{2}$. We may use law of sines,

$$\begin{aligned}\frac{AB}{\sin(\angle AQB)} &= \frac{BQ}{\sin(\angle BAQ)} \\ \Leftrightarrow BQ &= \frac{c \cdot \sin\left(A + \frac{B}{2} - 90\right)}{\sin\left(180 - \frac{B}{2}\right)} = \frac{c \cdot \sin\left(\frac{A}{2} - \frac{C}{2}\right)}{\sin\left(\frac{B}{2}\right)}\end{aligned}$$

However, we also know $\sin\left(\frac{B}{2}\right) = \frac{IE}{BI} = \frac{ID}{BI} = \frac{\sin(\angle IBD)}{\sin(\angle BDI)} = \frac{\sin\left(\frac{A-C}{2}\right)}{\sin\left(90 - \frac{A}{2}\right)}$.

$$\Leftrightarrow BQ = \frac{c \cdot \sin\left(\frac{A}{2} - \frac{C}{2}\right)}{\frac{\sin\left(\frac{A-C}{2}\right)}{\sin\left(90 - \frac{A}{2}\right)}} = c \cdot \sin\left(90 - \frac{A}{2}\right) = BP \quad \square$$

With the claim proven, note that,

$$\angle IBP = 180 - \angle AIB = 90 - \frac{C}{2} = \frac{A}{2} + \frac{B}{2}$$

Subtracting $\angle IBQ$ from this,

$$\begin{aligned} \angle PBQ &= \frac{A}{2} + \frac{B}{2} - A - \frac{B}{2} + 90 = 90 - \frac{A}{2} = \frac{B}{2} + \frac{C}{2} \\ \Leftrightarrow \angle BPQ &= \frac{180 - \angle PBQ}{2} = 90 - \frac{B}{4} - \frac{C}{4} \\ &\Leftrightarrow \frac{B}{4} + \frac{C}{4} \end{aligned}$$

We may angle chase to get $\angle PAX = \frac{3A}{4} + \angle PAB = \frac{3A}{4} = 90 + \angle \frac{A}{4}$ meaning the remaining angle $\angle AXP$ of the triangle has measure 45 as desired.

Problem 19 (Russian Killer)

Lemma: This works if $AB + CD = BC + AD$.

Proof: The quadrilateral inscribes a circle by Pitot. Where the touch points on AB, BC, CD, DA are W, X, Y, Z , we have,

$$\begin{aligned} r &= \frac{AW}{\cot\left(\frac{A}{2}\right)} + \dots + \frac{DZ}{\cot\left(\frac{D}{2}\right)} \\ &= \frac{AW + \dots + DZ}{\cot\left(\frac{A}{2}\right) + \dots + \cot\left(\frac{D}{2}\right)} \\ &= \frac{\frac{1}{2}(AB + \dots + DA)}{\cot\left(\frac{A}{2}\right) + \dots + \cot\left(\frac{D}{2}\right)} \end{aligned}$$

We have,

$$\begin{aligned} [ABCD] &= \left(\frac{1}{2} \cdot r \cdot AB\right) + \dots + \left(\frac{1}{2} \cdot r \cdot DA\right) \\ &= \frac{\frac{1}{4}(AB + \dots + DA)^2}{\cot\left(\frac{A}{2}\right) + \dots + \cot\left(\frac{D}{2}\right)} \quad \square \end{aligned}$$

We draw the angle bisectors of $\angle A, \dots, \angle D$ to form a quadrilateral $PQRS$ outside of $ABCD$. We slide A' along the segment of $PQRS$ which A lies on and define B', C', D' to be other points on the corresponding segments of $PQRS$ such that $AB \parallel A'B', \dots$ We slide A' until we get to a point such that $A'B' + C'D' = B'C' + D'A'$. This is possible since sliding A' in one direction, $A'D'$ and $B'C'$ increase while $A'B'$ and $D'C'$ decrease.

Let h be the distance from A' to AB . By congruent triangles, h is the distance from B' to BC and so on. Drawing the altitude from A', B' to AB , we get that,

$$A'B' = AB - h \tan \frac{A}{2} - h \tan \frac{B}{2}$$

$$C'D' = CD - h \tan \frac{C}{2} - h \tan \frac{D}{2}$$

$$B'C' = BC + h \tan \frac{B}{2} + h \tan \frac{C}{2}$$

$$D'A' = DA + h \tan \frac{D}{2} + h \tan \frac{A}{2}$$

Therefore, the perimeters of $ABCD$ and $A'B'C'D'$ are the same. Note that by Pitot's,

$$\begin{aligned} A'B' + C'D' &= B'C' + D'A' \\ \Leftrightarrow h &= \frac{AB + CD - BC - DA}{\tan \frac{A}{2} + \tan \frac{B}{2} + \tan \frac{C}{2} + \tan \frac{D}{2}} \end{aligned}$$

Therefore, since,

$$\begin{aligned} &[A'B'C'D'] - [ABCD] \\ &= h \cdot (A'B' - BC) + h \cdot (B'C' - BC) + h \cdot (C'D' - AD) + h \cdot (D'A' - AD) \\ &= h \cdot (AB + CD - BC - DA) \\ &= \frac{(AB + CD - BC - DA)^2}{\tan \frac{A}{2} + \tan \frac{B}{2} + \tan \frac{C}{2} + \tan \frac{D}{2}} \end{aligned}$$

We are done using the lemma.